



BAULKHAM HILLS HIGH SCHOOL

2014

HSC Assessment Task 1

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work
- Attempt all questions. Use the answer booklet provided.
- Answer each question on the appropriate page

Total marks –46

Exam consists of 5 pages.

This paper consists of TWO sections.

Section 1 – Pages 2-3

Multiple Choice

Question 1-5 (5 marks)

Section 2 – Pages 3-5

Extended Response

Question 6-10 (41 marks)

Section I - 5 marks**Answer in the table provided in the answer booklet**

1. The argument of iz where $z = 1 + i$ is

(A) $-\frac{\pi}{4}$

(B) $\frac{3\pi}{4}$

(C) $\frac{5\pi}{4}$

(D) $-\frac{3\pi}{4}$

2. If w is a non-real cube root of unity, the value of $\frac{1}{1+w^2} + \frac{1}{1+w}$ is equal to

(A) -1

(B) 0

(C) 1

(D) none of the above

3. $\arg(z) + \arg(\bar{z})$ equals

(A) 0

(B) $n\pi$

(C) $-n\pi$

(D) $\frac{\pi}{4}$

4. Consider the two statements below, for any complex number z with $|z| = 1$:

I. $z + \frac{1}{z} = 2\operatorname{Re}(z)$

II. $z - \frac{1}{z} = 2\operatorname{Im}(z)$

Which of these statements is always true?

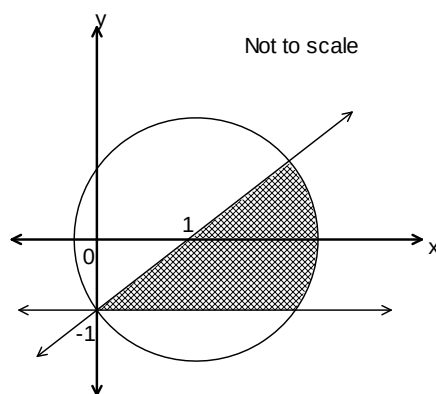
(A) Only statement I

(B) Only statement II

(C) Both statements

(D) Neither of the statements

5. The centre of the circle below is (1,0). Which inequalities define the shaded area?



- (A) $|z - 1| \leq \sqrt{2}$ and $0 \leq \arg(z - i) \leq \frac{\pi}{4}$
- (B) $|z - 1| \leq \sqrt{2}$ and $0 \leq \arg(z + i) \leq \frac{\pi}{4}$
- (C) $|z - 1| \leq 1$ and $0 \leq \arg(z - i) \leq \frac{\pi}{4}$
- (D) $|z - 1| \leq 1$ and $0 \leq \arg(z + i) \leq \frac{\pi}{4}$

End of Section I

Section II – Extended Response

Attempt questions 6-10.

Answer each question on the appropriate page of the answer booklet, showing all necessary working.

Question 6 (9 marks)

Marks

- (a) Let $z = 1 + i$ and $w = 4 - 2i$, find

(i) zw

1

(ii) $z + iw$

1

(iii) $\left| \frac{w}{z} \right|$

1

- (b) (i) Find all the complex numbers $z = a + ib$ such that $(a + ib)^2 = 5 - 12i$, where a, b are real numbers.

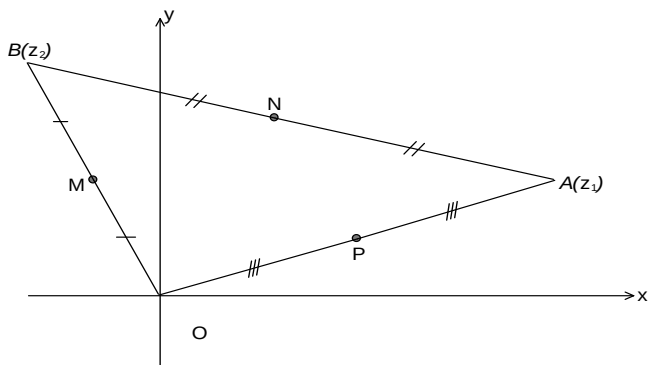
3

(ii) Hence solve $z^2 - (1 - 4i)z - (5 - i) = 0$

3

End of Question 6

Question 7 (9 marks)		Marks
a)	If $z = \sqrt{3} + i$ and $w = 1 - i$, (i) Write $\frac{z}{w}$ in the form $a+ib$ where a and b are real numbers. (ii) Write $\frac{z}{w}$ in mod-arg form (iii) Hence find the exact value of $\sin \frac{5\pi}{12}$	2 2 1
b)	z satisfies $z - 2i = 1$ and the point P represents z on an Argand diagram. (i) Sketch the locus of P (ii) Find the maximum and minimum values of $\arg z$, where $-\pi < \arg z < \pi$ (iii) Find the value of z when $\arg z$ takes its maximum value, expressing your answer in modulus-argument form	1 2 1
End of Question 7		

Question 8 (7 marks)		Marks
a)	w is a complex cube root of unity (ie w is a root of $z^3 = 1$) (i) Prove that w^2 is also a cube root of unity (ii) Evaluate $(1 + w)^3$, showing all working	1 2
b)	 $\triangle ABO$ lies on an Argand diagram. Points A and B represent the complex numbers z_1 and z_2 respectively. M, N and P are the midpoints of OB, AB and OA respectively. (i) Which complex number is represented by N? (ii) Express $\overrightarrow{AM}$ and $\overrightarrow{BP}$ in terms of z_1 and z_2 (iii) Hence simplify $\overrightarrow{ON} + \overrightarrow{AM} + \overrightarrow{BP}$	1 2 1
End of Question 8		

Question 9 (8 marks)		Marks
(i)	Solve $z^5 + 1 = 0$ by using De Moivre's theorem (Leave your answer in modulus-argument form)	2
(ii)	Explain why the solutions of $z^4 - z^3 + z^2 - z + 1 = 0$ are the non-real solutions of $z^5 + 1 = 0$	1
(iii)	Show that if $z^4 - z^3 + z^2 - z + 1 = 0$ where $z = \text{cis}\theta$, then $4 \cos^2 \theta - 2 \cos \theta - 1 = 0$	3
(iv)	Hence find the exact value of $\cos \frac{3\pi}{5}$	2
End of Question 9		

Question 10 (8 marks)		
a)	The region R in the Argand diagram is defined by $ z - 1 \leq z - i $ and $ z - 2 - 2i \leq 1$	
(i)	Sketch the region R	3
(ii)	If z lies on the boundary of region R, and $\arg(z - 1) = \frac{\pi}{4}$, find z in the form $a + ib$	3
b)	If $z = r\text{cis}\theta$, show that $\frac{z^2 - r^2}{z}$ is purely imaginary and state its value in terms of r and θ	2

End of Task

SECTION I.

1. $\arg z = \frac{\pi}{4}$

$\arg iz = \frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4}$ (B)

2. $\left. \begin{array}{l} w^3 = 1 \\ 1+w+w^2 = 0 \end{array} \right\} \frac{1}{1+w^2} + \frac{1}{1+w} = \left(\frac{1}{-w} + \frac{1}{-w^2} \right) \times w^3$
 $= -w^2 - w$
 $= -1$ (C)

3. 0 (A)

4. I is true

II is false (should be $2i \cdot \text{Im } z$)

$\therefore$ (A)

5. Circle: $r = \sqrt{2}$, centre $(1,0)$

$|z-1| \leq \sqrt{2}$

Arg: between 0 and $\frac{\pi}{4}$
 from $(0,-1)$

$0 \leq \arg(z+i) \leq \frac{\pi}{4}$

(B)

II.

Q6.

a) i) $zw = (1+i)(4-2i)$

$= 4 - 2i + 4i + 2$

$= 6 + 2i$

(1)

ii) $z+iw = 1+i + i(4-2i)$

$= 1+i + 4i + 2$

$= 3 + 5i$

(1)

iii) $\left| \frac{w}{z} \right| = \frac{|w|}{|z|} = \frac{\sqrt{20}}{\sqrt{2}} = \sqrt{10}$ (1)

b) i) $a^2 - b^2 + 2iab = 5 - 12i$ (1)

$a^2 - b^2 = 5$

$ab = -6$

$\left. \begin{array}{l} a^2 - b^2 = 5 \\ ab = -6 \end{array} \right\} (1) \therefore a = 3, b = -2 \text{ or } a = -3, b = 2$

$\therefore z = 3 - 2i, -3 + 2i$ (1)

ie. $z = \pm(3 - 2i)$

ii) $z = \frac{1-4i \pm \sqrt{(-(-1-4i))^2 - 4 \cdot 1 \cdot (-5-i)}}{2}$

$= \frac{1-4i \pm \sqrt{-15-8i + 20-4i}}{2}$

$= \frac{1-4i \pm \sqrt{5-12i}}{2}$ (1)

$= \frac{1-4i \pm (3-2i)}{2}$

$= \frac{4-6i}{2}, \frac{-2-2i}{2}$

$= 2-3i, -1-i$ (2) ← 1 each

Q7.

a) i) $\frac{z}{w} = \frac{\sqrt{3}+i}{1-i} \cdot \frac{1+i}{1+i}$ (1) rational den.

$$= \frac{\sqrt{3} + \sqrt{3}i + i - 1}{2}$$

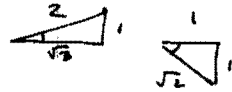
$$= \frac{\sqrt{3}-1}{2} + \frac{\sqrt{3}+1}{2}i \quad (1)$$

ii) $\frac{z}{w} = \frac{2 \operatorname{cis} \pi/6}{\sqrt{2} \operatorname{cis} -\pi/4}$ (1)

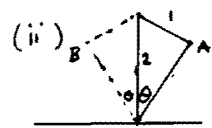
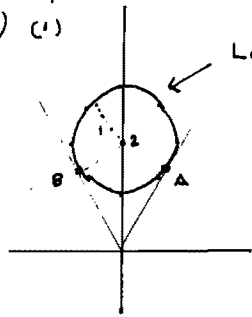
$$= \sqrt{2} \operatorname{cis} \frac{5\pi}{12} \quad (1)$$

iii) $\sqrt{2} \sin \frac{5\pi}{12} = \operatorname{Im}\left(\frac{z}{w}\right) = \frac{\sqrt{3}+1}{2}$

$$\therefore \sin \frac{5\pi}{12} = \frac{\sqrt{3}+1}{2\sqrt{2}} \quad (1)$$



b) i) $\text{Locus of } P(\text{circle}).$ (1)



$$\sin \theta = 1/2$$

$$\theta = \pi/6$$

$$\text{Max arg } z = \frac{\pi}{2} + \frac{\pi}{6} = \frac{2\pi}{3} \quad (\text{at } B)$$

$$\text{Min arg } z = \frac{\pi}{2} - \frac{\pi}{6} = \frac{\pi}{3} \quad (\text{at } A)$$

(iii) $OB = \sqrt{3}$ (by Pythagoras)

At B, $z = \sqrt{3} \operatorname{cis} \frac{2\pi}{3}$ (1)

Q8.

a) i) $\omega^3 = 1$ since ω is a root of $z^3 = 1$
 $(\omega^2)^3 = (\omega^3)^2 = 1^2 = 1$ (1)

$\therefore \omega^2$ is also a root.

(ii) $(1+\omega)^3 = (-\omega^2)^3$ (1) for $(-\omega^2)$

$$= -\omega^6$$

$$= -(\omega^3)^2$$

$$= -1$$

(1)

or $(1+\omega)^3 = 1 + 3\omega + 3\omega^2 + \omega^3$

$$= 1 + 3(\omega + \omega^2) + 1$$

(1) binomial + $\omega^3 = 1$

$$= 2 + 3(-1)$$

$$= -1.$$

(1)

b) i) $N = \frac{z_1 + z_2}{2}$ (1)

ii) $\vec{AM} = \frac{z_2}{2} - z_1$ (1)

$\vec{BP} = \frac{z_1}{2} - z_2$ (1)

iii) $\vec{ON} = \vec{AM} + \vec{BP}$

$$= \frac{z_1 + z_2}{2} + \frac{z_2}{2} - z_1 + \frac{z_1}{2} - z_2$$

$$= \frac{z_1 + z_2 + z_2 - 2z_1 + z_1 - 2z_2}{2}$$

$$= 0. \quad (1)$$

Q9.

i) Let $z = r \operatorname{cis} \theta$ and $z^5 = -1$

$$r^5 \operatorname{cis} 5\theta = \operatorname{cis} \pi \quad (\text{by De Moivre's theorem})$$

$$r^5 = 1 \quad \text{and} \quad 5\theta = \pi + 2k\pi \quad (k = \text{integer})$$

$$r = 1 \quad \theta = \frac{(2k+1)\pi}{5} \quad \text{--- (1)}$$

$$\therefore z = \operatorname{cis} \frac{\pi}{5}, \operatorname{cis} \frac{3\pi}{5}, \operatorname{cis} \pi, \operatorname{cis} \frac{7\pi}{5}, \operatorname{cis} \frac{9\pi}{5}$$

or

$$z = \operatorname{cis} \left(\pm \frac{\pi}{5} \right), \operatorname{cis} \left(\pm \frac{3\pi}{5} \right), -1. \quad \text{--- (1)}$$

ii) $z^5 + 1 = (z+1)(z^4 - z^3 + z^2 - z + 1) = 0$

when

$$z+1=0$$

$$z = -1$$

(the real soln of $z^5 + 1 = 0$)

$$\text{or } \underbrace{z^4 - z^3 + z^2 - z + 1 = 0}_{\text{The solns will be the non-real solns of } z^5 + 1 = 0.}$$

(1)

iii) If $z^4 - z^3 + z^2 - z + 1 = 0$

(divide by z^2)

$$\text{then } z^2 - z + 1 - \frac{1}{z} + \frac{1}{z^2} = 0$$

$$\left(z^2 + \frac{1}{z^2} \right) - \left(z + \frac{1}{z} \right) + 1 = 0 \quad (1)$$

Since $|z| = 1$: $z + \frac{1}{z} = 2 \cos \theta$, $z^2 + \frac{1}{z^2} = 2 \cos 2\theta$ (1)

$$\therefore 2 \cos 2\theta - 2 \cos \theta + 1 = 0$$

$$2(2 \cos^2 \theta - 1) - 2 \cos \theta + 1 = 0 \quad (1)$$

$$\underline{4 \cos^2 \theta - 2 \cos \theta - 1 = 0}$$

iv) $z = \operatorname{cis} \frac{3\pi}{5}$ is one of the solutions of $z^4 - z^3 + z^2 - z + 1 = 0$ (from ii)

$$\therefore \theta = \frac{3\pi}{5} \text{ is one of the solns of } 4 \cos^2 \theta - 2 \cos \theta - 1 = 0$$

$$\therefore \cos \theta = \frac{2 \pm \sqrt{4 - 4 \cdot 4 \cdot (-1)}}{8}$$

$$= \frac{2 \pm \sqrt{20}}{8} \quad (1)$$

$$= \frac{2(1 \pm \sqrt{5})}{8}$$

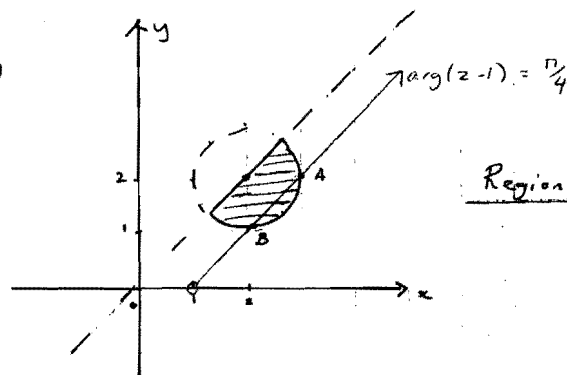
$$= \frac{1 \pm \sqrt{5}}{4}$$

Now $\frac{3\pi}{5}$ is obtuse $\therefore \cos \frac{3\pi}{5} < 0$

$$\therefore \cos \frac{3\pi}{5} = \frac{1 - \sqrt{5}}{4} \quad (1)$$

Q10.

a) (i)



Region R shaded.

- (1) - circle
- (1) - line
- (1) - shading

$$(ii) \quad \arg(z-1) = \frac{\pi}{4} \Rightarrow y = x - 1 \quad \text{in Cartesian form (1)}$$

$$|z - 2 - 2i| = 1 \Rightarrow (x-2)^2 + (y-2)^2 = 1 \quad \text{... (2)}$$

z is given by the two points of intersection A and B.

$$(x-2)^2 + (x-3)^2 = 1$$

$$x^2 - 4x + 4 + x^2 - 6x + 9 = 1$$

$$2x^2 - 10x + 12 = 0 \quad (1)$$

$$2(x^2 - 5x + 6) = 0$$

$$2(x-3)(x-2) = 0$$

$$x = 3 \quad x = 2$$

$$y = 2 \quad y = 1$$

$$\therefore z = 3 + 2i, \quad 2 + i$$

$$b) \quad \frac{z^2 - r^2}{z} = \frac{z^2 - z\bar{z}}{z}$$

$$= z - \bar{z}$$

$$= 2i \operatorname{Im}(z)$$

$$= 2i \cdot r \sin \theta$$

(1) ... use $z\bar{z}$

(1)

or

$$z - \frac{z\bar{z}}{z} = z - \bar{z}$$

$$= 2i \operatorname{Im}(z)$$

$$= 2i \cdot r \sin \theta$$

(1) use $z\bar{z}$

(1)

or

$$\frac{z^2 - r^2}{z} = \frac{r^2 \operatorname{cis} 2\theta - r^2}{r \operatorname{cis} \theta}$$

$$= \frac{r^2 (\operatorname{cis} 2\theta - 1)}{r \operatorname{cis} \theta}$$

$$= r \left(\frac{\operatorname{cis} 2\theta}{\operatorname{cis} \theta} - \frac{1}{\operatorname{cis} \theta} \right)$$

$$= r (\operatorname{cis} \theta - \operatorname{cis}(-\theta)) \quad \dots (1)$$

$$= r (\cos \theta + i \sin \theta - (\cos \theta - i \sin \theta))$$

$$= r (\cancel{\cos \theta} + i \sin \theta - \cancel{\cos \theta} + i \sin \theta)$$

$$= r \cdot 2i \sin \theta$$

$$= 2i \cdot r \sin \theta$$

(1)